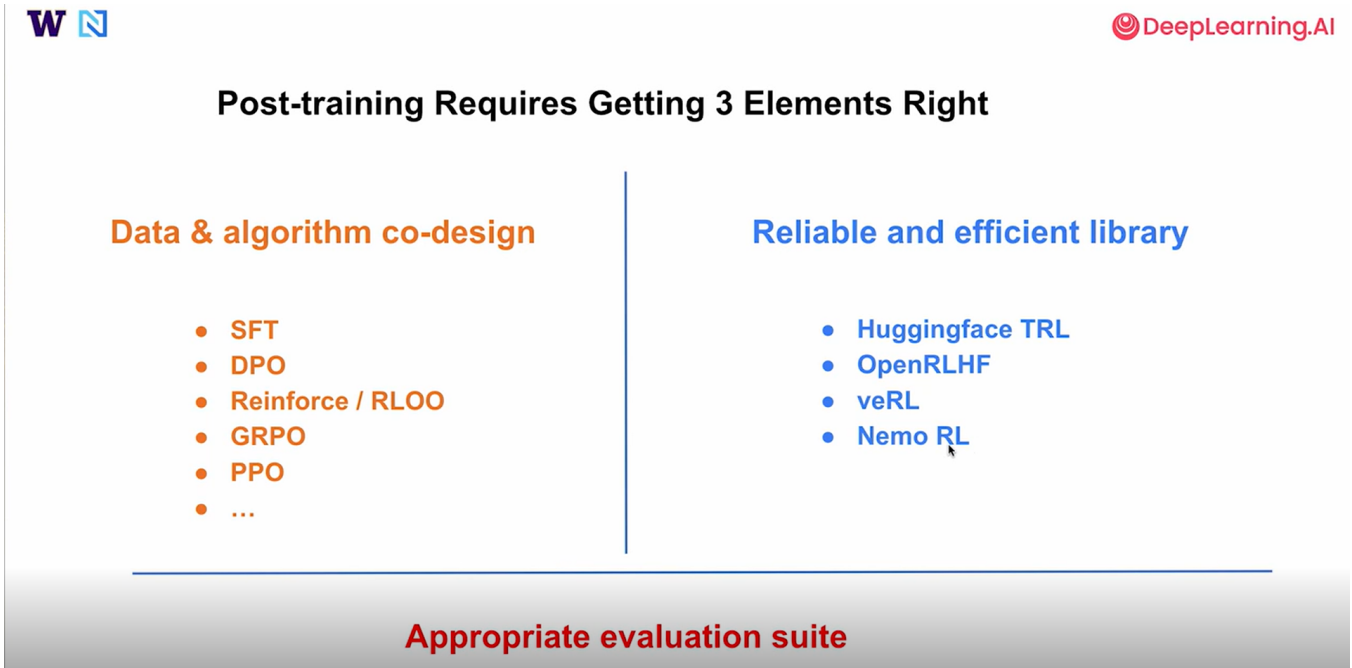


# Post-training Requires Getting 3 Elements Right

The fourth screenshot highlights three practical requirements for successful LLM post-training: data & algorithm co-design, a reliable/efficient training library, and an appropriate evaluation suite.



## 22. The big idea: post-training is not just about choosing an algorithm

The slide says post-training requires getting **three elements right**. In practice, a strong post-training system needs the right **training data and algorithm together**, a **reliable implementation stack**, and an **evaluation suite** that can tell whether the changes actually improved the model.

## 23. Element 1 — Data & algorithm co-design

This means the training method and the training data should be designed together. The same dataset is not automatically appropriate for every post-training method. The supervision signal must match the algorithm.

**SFT:** needs high-quality demonstrations such as prompt → ideal response.

**DPO:** needs preference data such as prompt → chosen response + rejected response.

**Reinforce / RLOO:** needs prompts and a reward signal that can score generated responses.

**GRPO:** commonly uses groups of sampled responses and relative/group-based reward signals, depending on the implementation.

**PPO:** uses a reinforcement-learning setup with rewards and policy/value-related machinery.

**Key lesson:** changing the algorithm without considering the data format, quality, reward signal, or task can produce poor training results.

## 24. Methods shown on the slide

| Method           | Core training signal / idea                                                                                                                               |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| SFT              | Learn from curated demonstrations: prompt → desired response.                                                                                             |
| DPO              | Learn preference between chosen and rejected responses.                                                                                                   |
| Reinforce / RLOO | Generate responses, obtain rewards, and optimize the policy toward higher reward; RLOO is a policy-gradient variant using leave-one-out reward baselines. |